



Cesar Augusto VALADES CRUZ, PhD

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cessvala.github.io

Researcher with extensive experience in the development and operation of advanced optical microscopes, and in the processing and analysis of imaging data applied to cell biology and microbiology. His work integrates machine learning, large-scale data handling, high-content imaging, and 3D+time data visualization for the study of cellular processes. His research is organized around three main lines: (1) development and application of super-resolution microscopy, light-sheet microscopy, and nanoscale molecular orientation analysis; (2) 3D+time tracking of intracellular events; and (3) development of algorithms for large-scale biological image analysis.

PROFESSIONAL EXPERIENCE

Associate researcher — Algal Growth and Development TEAM of Prof. Zhang Cheng-Cai (02/2023 – Current)

Institute of Hydrobiology, Chinese Academy of Sciences, Wuhan, China

Research focuses on the study of the growth and differentiation of cyanobacterial cells using advanced microscopy and deep learning algorithms.

Postdoctoral researcher — Team SERPICO Paris of Dr. C. Kervrann and Dr. J. Salamero (01/2022 – 01/2023)

French National Institute for Research in Digital Science and Technology (Inria) & CNRS - Curie Institute, Paris, France

Project: Polarization Microscopy for Imaging of Membrane Organization (PoMIMO). LabEx Cell(n)Scale collaborative project between Inria and Curie Institute. Implementation of polarization module in a STORM-TIRFM (PolarSTORM) system at Curie Institute. Developing of image analysis workflow of biomolecules tracking and estimating spatial high-resolution maps of molecular mobility.

Postdoc/Engineer — Inria and startup Biotech Myriade (01/2021 – 12/2021)

Project: Improvement of tracking and size estimation of virus and extracellular vesicles. Collaboration between SERPICO Team and the Biotech startup Myriade. I develop mathematical models and image processing tools to improve the tracking and size estimation of viruses and extracellular vesicles in VideoDrop.

Inria Starting Research Position — Team SERPICO Paris (04/2019 – 12/2020)

French National Institute for Research in Digital Science and Technology (Inria) & CNRS – Curie Institute, Paris, France

Project: Acquisition, analysis, and visualization of 3D Dynamic cellular imaging of endocytosis/recycling mechanisms in the membrane during cell migration using machine learning. I work in novel machine learning methods of image processing able to detect the main regions of interest, and automatic quantification of molecular interactions and cell processes. In addition, I collaborate in the development of machine learning-driven navigation and interaction techniques for 3D+Time data enabling the analysis of localized intra-cellular events (endocytosis and exocytosis) and cell processes (migration, division, etc.).

Postdoctoral researcher — Team of Prof. Ludger JOHANNES, Curie Institute, Paris, France (01/2016 – 03/2019)

Project: Advanced cellular imaging of endocytosis. Development of 3D image processing and quantification methods to study different modes of endocytosis, using advanced high spatio-temporal resolution imaging and single particle tracking. In addition, I was also responsible of setting up a Lattice Light Sheet Microscope on the PICT-IBiSA imaging facility at Curie Institute, which is part of the France BioImaging National Infrastructure.

University Lecturer — Monterrey Institute of Technology and Higher Education, Mexico (01/2015 – 12/2015)

Teaching Physics, mathematics, and differential equations.

Research Engineer, CNRS — Fresnel Institute, Marseille, France (12/2013 – 04/2014)

Responsible of developing image processing and quantification methods for super-resolution microscopy (dSTORM).

Ph.D with Dr. Sophie BRASSELET and Dr. Pablo LOZA-ALVAREZ — Fresnel Institute, Marseille, France & Institute of Photonic Sciences, Barcelona, Spain (12/2010 – 07/2014)

Thesis: Polarized Super-Resolution Fluorescence Microscopy. Implementation of a novel method of super resolution microscopy, in combination with a polarized detection to study molecular orientation behaviors, to report structural information at the single molecule and at nanometric spatial scale.

EDUCATION

PhD, Photonics Engineering, Nanophotonics & Biophotonics — Aix-Marseille University, France & Polytechnic University of Catalonia, Spain (2010 – 2014)

- Erasmus Mundus program. Thesis on polarized super-resolution fluorescence microscopy — a new way to visualize molecular orientation at the nanoscale. Awarded Cum Laude.

MSc, Biophotonics for Telecommunications and Biotechnologies — Ecole Normale Supérieure de Cachan, France & Complutense University of Madrid, Spain (2008 – 2010)

- Erasmus Mundus program. Awarded Magna Cum Laude.

Master of Engineering, Quality Systems and Productivity — Monterrey Institute of Technology and Higher Education, Mexico (2009 – 2011)

- Focus on lean manufacturing and quality management systems.

B.S., Mechatronics Engineering — Monterrey Institute of Technology and Higher Education, Mexico (2003 – 2008)

- Graduated with Honors.

SKILLS

Computational Languages & Tools: MATLAB, Python, C/C++, LabVIEW, Java, ImageJ/FIJI, Icy, IMARIS, GPU programming, Parallel computing, TensorFlow, Keras, Machine Learning, R, Prism, Microsoft Excel, Zen celldiscover, IDEAS (ImageStream)

PROJECTS & COLLABORATIONS

2024 National Natural Science Foundation of China (NSFC) Special Project (32350028): Establishing cell function editing technology based on endosymbiosis.

2024 National Natural Science Foundation of China (NSFC) Key Project (32330003): Three-dimensional regulatory mechanisms of cell division in multicellular cyanobacteria and their relationship with morphological diversity and peptidoglycan metabolism.

2022 Project POMIMO: Polarization Microscopy for Imaging of Membrane Organization. Create a new imaging approach that will have the potential to be applied to the topics where the spatial and molecular organization of lipids/proteins defines both the structure and function of assemblies in reconstituted systems and in living cells.

2019 Project NAVISCOPE: image-guided navigation and visualization of large data sets in live cell imaging and microscopy. INRIA IPL project, initiated to implement novel machine-learning methods able to detect the main regions of interest, and automatic quantification of sparse sets of molecular interactions and cell processes during navigation to save memory and computational resources.

2019 Project BiolmageIT: open-source integrator for Image DATA management and analysis. Project of the Serpico TEAM in the frame of the NRI (National Research Infrastructure – France BioImaging) and dissemination toward the 18 Imaging Facilities that constitute the Core of the Infrastructure.

2019 Project: Ultrastructure imaging of actin assemblies imaged by polarized light sheet microscopy. Ongoing collaboration in the frame of France BioImaging R&D program for image processing of polarized light sheet microscopy data with Dr. Sophie Brasselet, Fresnel Institute.

2017 Project ANR: Data Assimilation and Lattice Light Sheet imaging for endocytosis/exocytosis pathway modeling in the whole cell (DALLISH). Collaboration to investigate endocytosis pathways in the whole cell using 3D single particle tracking.

FELLOWSHIPS & DISTINCTIONS

- 2025–2027 Member, Scientific Advisory Committee, GloBIAS (Global Bioimage Analysis Society)
- 2024–2027 President's International Fellowship Initiative (PIFI) Special Expert, Chinese Academy of Sciences
- 2020–2024 Member of the Mexican National Research System (SNI 1)
- 2010–2014 Erasmus Mundus Fellowship for PhD
- 2008–2010 Mexican National Council of Science and Technology (CONACyT) fellowship for Master
- 2008–2010 Erasmus Mundus Fellowship for Master

First author publications

- [1] Senthil Kumar C. S.*, **Valades-Cruz C. A.***, Sison M.*, Vesga A. G., Rey-Barroso J., Curcio V., Aleman-Castaneda L. A., Alonso M. A., Poincloux R., Mavrikakis M., Brasselet S. 4polar3D single molecule imaging of 3D orientation in dense actin networks using ratiometric polarization splitting. *Nature Communications* (2026)
- [2] **Valades-Cruz C. A.***, Barth R.*, Abdellah M.*, Shaban H. A. Genome-wide analysis of the biophysical properties of chromatin and nuclear proteins in living cells with Hi-D. *Nature Protocols* (2025)
- [3] Papereux S.*, Leconte L.*, **Valades-Cruz C. A.***, Liu T., Dumont J., Chen Z., Salamero J., Kervrann C., Badoual A. DeepCristae, a CNN for the restoration of mitochondria cristae in live microscopy images. *Communications Biology* (2025)
- [4] Prigent S.*, **Valades-Cruz C.A.***, Leconte L.*, Maury L., Salamero J., Kervrann C. BiolmageIT: Open-source framework for integration of image data-management with analysis. *Nature Methods* (2022)
- [5] Vaz Rimoli C.*, **Valades-Cruz C.A.***, Curcio V., Mavrikakis M., Brasselet S. 4polar-STORM polarized super-resolution imaging of actin filament organization in cells. *Nature Communications* (2022)
- [6] Shaban H.*, **Valades-Cruz C. A.***, Savatier J., Brasselet S. Polarized super-resolution structural imaging inside amyloid fibrils using Thioflavine T. *Scientific Reports* (2017)
- [7] **Valades-Cruz C. A.***, Shaban H.*, Kress A., Bertaux N., Monneret S., Mavrikakis M., Savatier J., Brasselet S. Quantitative nanoscale imaging of orientational order in biological filaments by polarized superresolution microscopy. *PNAS* (2016)

Corresponding author publications

- [1] Prigent S.[†], **Valades-Cruz C.A.[†]**, Leconte L., Salamero J., Kervrann C. STracking: a free and open-source python library for particle tracking and analysis. *Bioinformatics* (2022)

[†] Co-corresponding authors.

Other publications

- [1] Masson A., Prigent S., **Valades-Cruz C. A.**, Leconte L., Maury L., Salamero J., Kervrann C. BiolmageIT: A novel python-based architecture for reproducible bio-image workflows. *Journal of Microscopy* (2026)
- [2] Shafaq-Zadah M., Dransart E., Hamitouche I., Wunder C., Chambon V., **Valades-Cruz C. A.**, Leconte L., Sarangi N. K., Robinson J., Bai S-K., Regmi R., Di Cicco A., Hovasse A., Bartels R., Nilsson U.J., Cianferani-Sanglier S., Leffler H., Keyes T. E., Levy D., Raunser S., Roderer D., Johannes L. Spatial N-glycan rearrangement on $\alpha 5\beta 1$ integrin nucleates galectin-3 oligomers to determine endocytic fate. *Nature Communications* (2025)
- [3] MacDonald E., Forrester A., **Valades-Cruz C. A.**, Madsen T. D., Hetmanski J.H.R., Dransart E., Ng Y., Godbole R., Shp A. A., Leconte L., Chambon V., Ghosh D., Pinet A., Bhatia D., Lombard B., Loew D., Larsen M. R., Leffler H., Lefeber D.J., Clausen H., Blangy A., Caswell P., Shafaq-Zadah M., Mayor S., Weigert R., Wunder C., Johannes L. Growth factor-triggered de-sialylation controls glycolipid-lectin-driven endocytosis. *Nature Cell Biology* (2025)
- [4] Li S-Y., He C., **Valades-Cruz C. A.**, Zhang C-C., Yang Y. Phototactic signaling network in rod-shaped cyanobacteria: a study on *Synechococcus elongatus* UTEX 3055. *Microbiological Research* (2025)
- [5] Lemaigre C., Ceuppens A., **Valades-Cruz C. A.**, Ledoux B., Vanbeneden B., Hassan M., Zetterberg F. R., Nilsson U. J., Johannes L., Wunder C., Renard H-F., Morsomme P. N-BAR and F-BAR proteins – Endophilin-A3 and PSTPIP1 – control clathrin-independent endocytosis of L1CAM. *Traffic* (2023)
- [6] Prigent S., Nguyen H-N., Leconte L., **Valades-Cruz C. A.**, Hajj B., Salamero J., Kervrann C. SPITFIR(e): a supermaneuverable algorithm for fast denoising and deconvolution of 3D fluorescence microscopy images and videos. *Scientific Reports* (2023)
- [7] Forrester A., Rathjen S., Garcia-Castillo M. D., Bachert C., Couhert A., Tepshi L., Pichard S., Martinez J., Munier M., Sierocki R., Renard H.F., **Valades-Cruz C.A.**, Dingli F., Loew D., Lamaze C., Cintrat J.C., Linstedt A., Gillet D., Barbier J., Johannes L. Functional Dissection of the Retrograde Shiga Toxin Trafficking Inhibitor Retro-2. *Nature Chemical Biology* (2020)
- [8] Renard H-F, Tyckaert F., Lo Giudice C., Hirsch T., **Valades-Cruz C.A.**, Lemaigre C., Shafaq-Zadah M., Wunder C., Wattiez R., Johannes L., van der Bruggen P., Alsteens D., Morsomme P. Endophilin-A3 and Galectin-8 control the clathrin-independent endocytosis of CD166. *Nature Communications* (2020)
- [9] Briane V, Vimond M, **Valades-Cruz CA**, Salomon A, Wunder C, Kervrann C. A sequential algorithm to detect diffusion switching along intracellular particle trajectories. *Bioinformatics* (2020)
- [10] Salomon A., **Valades-Cruz C. A.**, Leconte L., Kervrann C. Dense Mapping of Intracellular Diffusion and Drift from Single-Particle Tracking Data Analysis, ICASSP 2020 - 2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Barcelona, Spain, 2020, pp. 966-970.
- [11] Torrino S., Shen W., Blouin C., Kailasam Mani S., Viaris de Lesegno C., Bost P., Grassard A., Koster D., **Valades-Cruz C. A.**, Chambon V., Johannes L., Pierobon P., Soumelis V., Coirault C., Vassilopoulos S., Lamaze C. EHD2 is a mechanotransducer connecting caveolae dynamics with gene transcription. *J Cell Biol.* (2018)
- [12] Banerjee A., Grazon C., Pons T., Bhatia D., **Valades-Cruz C. A.**, Johannes L., Krishnan Y., Dubertret B. A Novel Type of Quantum Dot–Transferrin Conjugate Using DNA Hybridization Mimics Intracellular Recycling of Endogenous Transferrin. *Nanoscale* (2018)

Reviews, Perspectives & Comments

- [1] Ambroset M., **Valades-Cruz C. A.**, Tonfack L. B., D'Antuono R., Fuster-Barcelo C. Assessing the reproducibility of a bioimage analysis workflow characterising tissue flow in *Drosophila*. *Journal of Microscopy* (2026)
- [2] **Valades-Cruz C. A.** et al. Challenges of intracellular visualization using virtual and augmented reality. *Front. Bioinform.* (2022)
- [3] Johannes L., **Valades-Cruz C. A.** The final twist in endocytic membrane scission. *Nature Cell Biology* (2021)

Review assignments journals: *Nat. Communications, Bioinformatics, PLOS Comput. Biol., J. Phys. Chem. Lett. & J. Vis. Exp.*